Strategic Product Placement Analysis

1-INTRODUCTION:

This project aims to investigate the relationship between product positioning, sales performance, and consumer behavior. Using Tableau, we will analyze data to uncover insights into how different positioning strategies impact sales and consumer preferences. By visualizing the data, we aim to provide actionable recommendations to optimize product positioning strategies and drive revenue growth.

A retail company wants to understand the impact of product positioning on its sales and consumer behavior. They have collected data on sales figures, product placement, and consumer demographics. They seek insights into which product positioning strategies are most effective in driving sales and how they can tailor their marketing efforts accordingly. Through data visualization with Tableau, the company hopes to gain actionable insights to improve its product positioning strategies and increase revenue.

1.1-Project Overview:

This project focuses on analyzing how product positioning influences sales performance and consumer behavior using data visualization techniques. The objective is to examine product-related data, identify market trends, and evaluate the effectiveness of different positioning strategies through interactive dashboards developed in Tableau.

The project involves collecting and preprocessing the dataset, cleaning and filtering the data, and creating meaningful visualizations such as sales trends, product category performance, customer ratings, pricing analysis, and market comparisons. These visualizations help uncover patterns in customer preferences, product demand, and business performance.

By analyzing the relationship between product positioning, pricing, sales volume, customer ratings, and market factors, the project provides actionable insights that can support data-driven business decisions. The findings can help organizations optimize product placement, improve marketing strategies, enhance customer satisfaction, and increase overall sales performance.

Tools C Technologies Used:

1. Tableau – Data visualization and dashboard creation.
2. Microsoft Excel/CSV– Data preparation and preprocessing.
3. Data Analytics Techniques– Data cleaning, filtering, aggregation, and trend analysis.

1.2-Purpose:

The purpose of this project is to analyze the impact of product positioning on sales performance and consumer behavior using data visualization techniques. By examining product data through Tableau, the project aims to identify market trends, customer preferences, and factors that influence product success.

The project helps businesses:

* Understand how different product positioning strategies affect sales.

* Identify high-performing and low-performing products.

* Analyze customer ratings, pricing, and sales trends.

* Discover insights into consumer purchasing behavior.

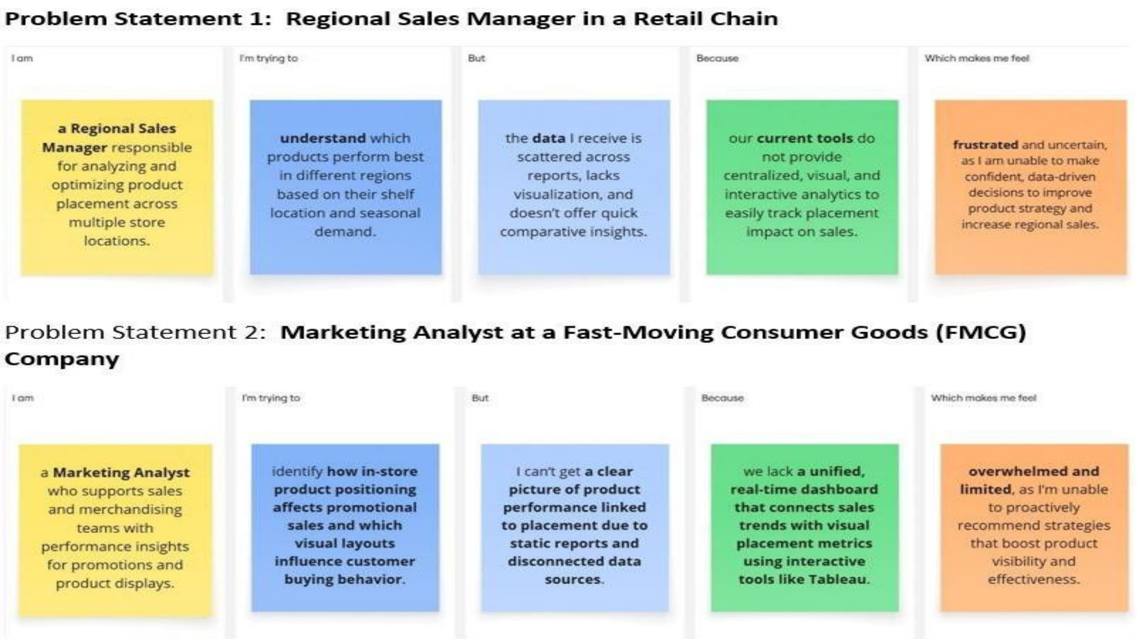
* Support data-driven decision-making for marketing and product strategies.

* Improve product positioning to increase customer satisfaction, sales, and overall business profitability.

2- IDEATION PHASE:

The ideation phase focuses on generating and evaluating ideas to solve the identified business problem. Different approaches are explored to analyze product positioning, sales performance, and consumer behavior. The most effective solution is selected based on feasibility and business value. This phase provides the foundation for
designing the Tableau dashboards and analytics workflow.

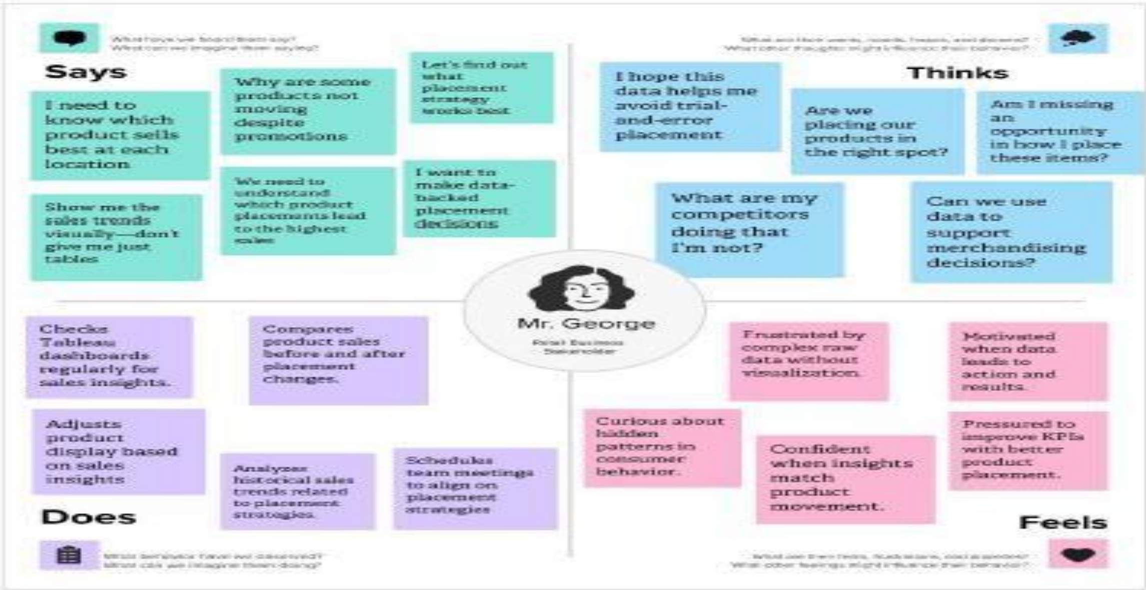
2.1-Problem Statement:



Problem Statement (PS)	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1: Regional Sales Manager in a Retail Chain	a Regional Sales Manager responsible e for analyzing and optimizing product placement across multiple store locations.	understand which products perform best in different regions based on their shelf location and seasonal demand.	the data I receive is scattered across reports, lacks visualized ion, and doesn't offer quick compara tive insights.	our current tools do not provide centralized, visual, and interactive analytics to easily track placement impact on sales.	frustrated and uncertain, as I am unable to make confident, data driven decisions to improve product strategy and increase regional sales.
PS-2: Marketing Analyst at a FastMoving Consumer Goods (FMCG) Company	a Marketing Analyst who supports sales and merchandi sing teams with performan ce insights for promotions and product displays.	identify how instore product positioning affects promotional sales and which visual layouts influence customer buying behavior.	I can't get a clear picture of product performa nce linked to placeme nt due to static reports and disconne cted data sources.	we lack a unified, real-time dashboard that connects sales trends with visual placement metrics using interactive tools like Tableau.	overwhelmed and limited, as I'm unable to proactively recommend strategies that boost product visibility and effectiveness.

2.2 -Empathy Map Canvas :

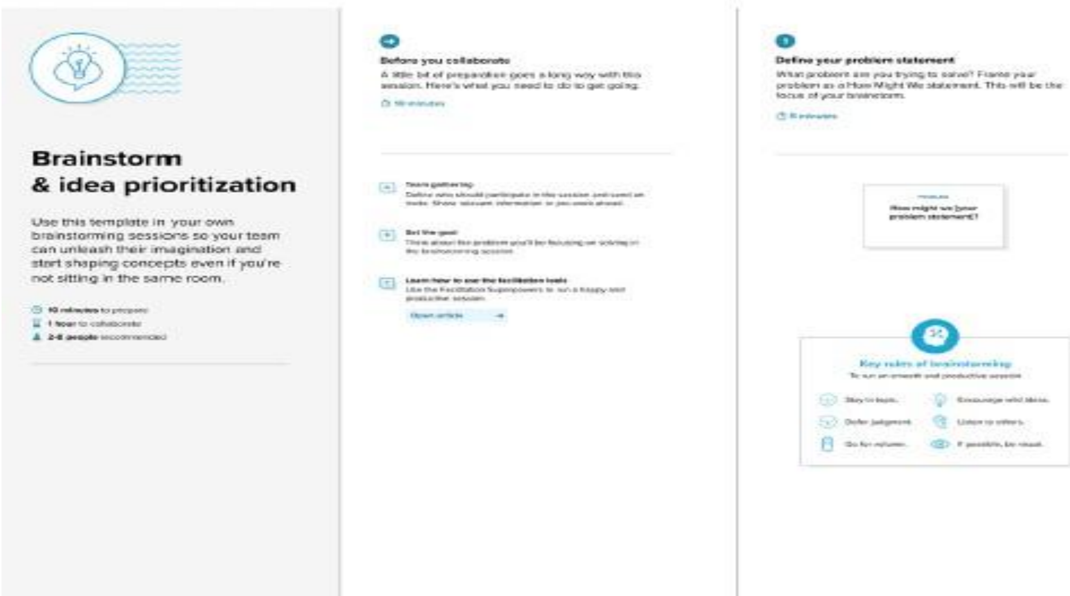
Empthy Map-1:Strategic Product Placement Analysis



Empthy Map-2:Strategic Product Placement Analysis

2.3- Brainstorming- Idea Generation- Prioritizaation Template:

Step-1: Team Gathering, Collaboration and Select the Problem Statement



3- REQUIREMENT ANALYSIS:

Requirements analysis is the process of identifying and understanding the needs and expectations of stakeholders for a system or project.It helps define clear, complete, and accurate requirements before development begins.This process reduces errors, misunderstandings, and costly changes during the project.The final output is a well-documented set of requirements that guides the design and development of the system.

3.1-Customer Journey map:

3.2-Solution Requirement:

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Gathering	Collect sales, price, promotion, and product placement data from retail databases.
FR-2	Data Cleaning C Preparation	Remove duplicates, handle missing values, and normalize data for consistency.
FR-3	Visualizations	Create Unique Visualizations , for example: 1. Avg Sales Volume vs Product Category 2. Avg Sales Volume by Product Category by Product Position 3. Avg Sales Volume by Product Category by Season 4. Competitor Price Vs Price 5. Consumer Demographics vs Sales Volume 6. Foot Traffic by Avg Sales Volume 7. Product Category vs Price 8. Promotion of Product Category on Price and Sales Volume
FR-4	Dashboard Development	Create interactive dashboards in Tableau.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

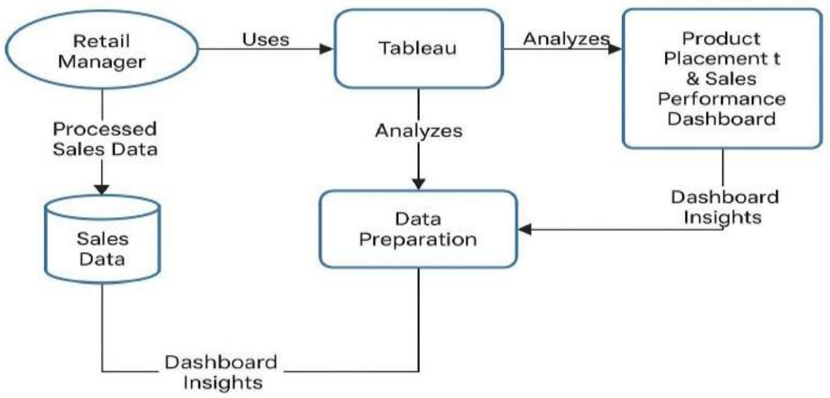
FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The dashboard must be user-friendly with intuitive navigation, filters, and clear visualizations.
NFR-2	**Security**	Ensure role-based access, data encryption, and secure connections (SSL/HTTPS).
NFR-3	**Reliability**	The solution should consistently deliver accurate and updated insights without failure.
NFR-4	**Performance**	Dashboards must load within 5 seconds and support quick interactions with filters.
NFR-5	**Availability**	The dashboard should be available 24/7 with minimal downtime (<1% monthly).
NFR-6	**Scalability**	The system must handle growing data volumes and users without degrading performance.

3.3-Data Flow Diagram:

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Data Analyst	Data Collection	USN1	Gathering Data	Required data is collected successfully from all sources	High	Sprint 1
Data Analyst	Data Collection	USN2	Loading Data	Data is loaded into Tableau without errors	High	Sprint 1
Data Analyst	Data Preparation	USN3	Handling Missing Values	Missing values are identified and handled correctly	High	Sprint 1
Data Analyst	Data Preparation	USN4	Creating Fields	Required calculated/custom fields are created	High	Sprint 1
Data Analyst	Data Preparation	USN5	Handling Inconsistency in Data	Inconsistent values are cleaned and standardized	High	Sprint 1
Business User	Data Visualization	USN6	Create Bar Chart	Bar chart displays correct comparison of data	Medium	Sprint 2
Business User	Data Visualization	USN7	Create Pie Chart	chart correctly shows category distribution	Medium	Sprint 2
Business User	Data Visualization	USN8	Create Line Chart	chart correctly displays trends over time	Medium	Sprint 2
Business User	Data Visualization	USN9	Create Map Visualization	Map displays geographical data accurately	Medium	Sprint 2
Business User	Dashboard	USN10	Develop Dashboard	Interactive dashboard with filters and visualizations is completed	High	Sprint 2
Business User	Story	USN11	Tableau Story	presents insights in the correct sequence	High	Sprint 2

3.4-Technology Stack:

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

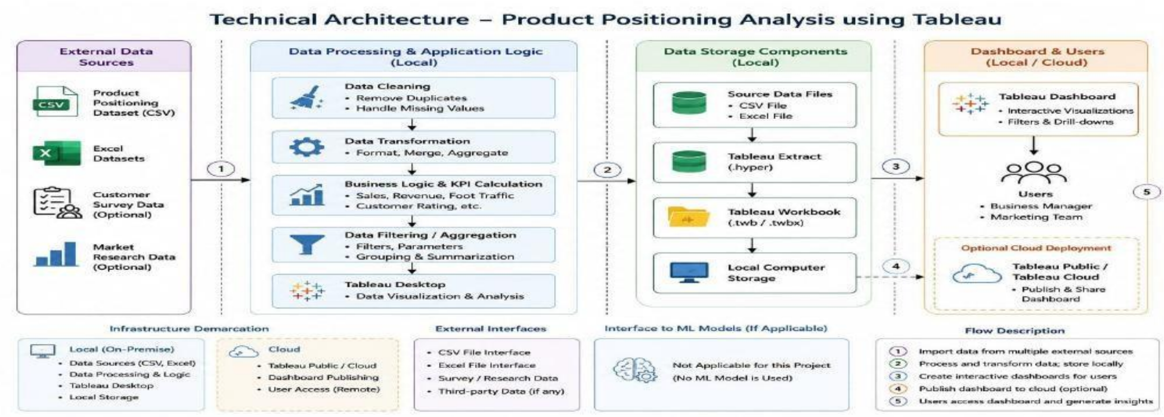


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	Data Source	Raw sales data collected from POS,ERP, or CRM systems	Microsoft SQL Server / Excel / CSV
2.	Data Preparation	Cleans, transforms, and loads data into Tableau	Tableau Prep / Python / SQL
3.	Data Visualization Tool	Creates dashboards and interactive visual reports	Tableau Desktop / Tableau Online
4.	User Interface	Web or embedded view where users interact with dashboards	Tableau Public / Embedded Tableau
5.	Access Control	Manages user roles and access levels	Tableau Server / Active Directory
6.	Report Sharing	Mechanism to export and distribute dashboard insights	Tableau Share / PDF Export / Email
7.	Feedback Collection	Collects feedback from users about the dashboard experience	Google Forms / Microsoft Forms

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1.	Usability	Easy-to-use dashboard with filters, tooltips, and clean layout	Tableau UX / UI design best practices
2.	Performance	Fast data loading and interaction with filters	Tableau Extracts (TDE/Hyper), Optimized SQL
3.	Security	Role-based access and secure data handling	Tableau Server Permissions / HTTPS

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3.	Security	Role-based access and secure data handling	Tableau Server Permissions / HTTPS
4.	Scalability	Ability to handle large datasets and user growth	Tableau Server / Cloud Storage
5.	Availability	Ensures the dashboard is accessible 24/7 with minimal downtime	Cloud Hosting (Azure/AWS), Tableau Server
6.	Integration	Seamless connection to different data sources	Tableau Connectors / APIs
7.	Real-time Data Refresh	Automatically updates dashboards with latest data	Tableau Scheduler / Live Data Connection

4- PROJECT DESIGN:

The project design defines how data is collected, cleaned, and organized before creating visualizations in Tableau.It identifies the key business objectives and performance metrics to be analyzed.Interactive dashboards and charts are designed to present insights in a clear and user-friendly manner.Filters, parameters, and calculated fields are used to enable dynamic data exploration.The final design ensures accurate reporting, easy navigation, and effective decision-making.

4.1-Problem Solution Fit:

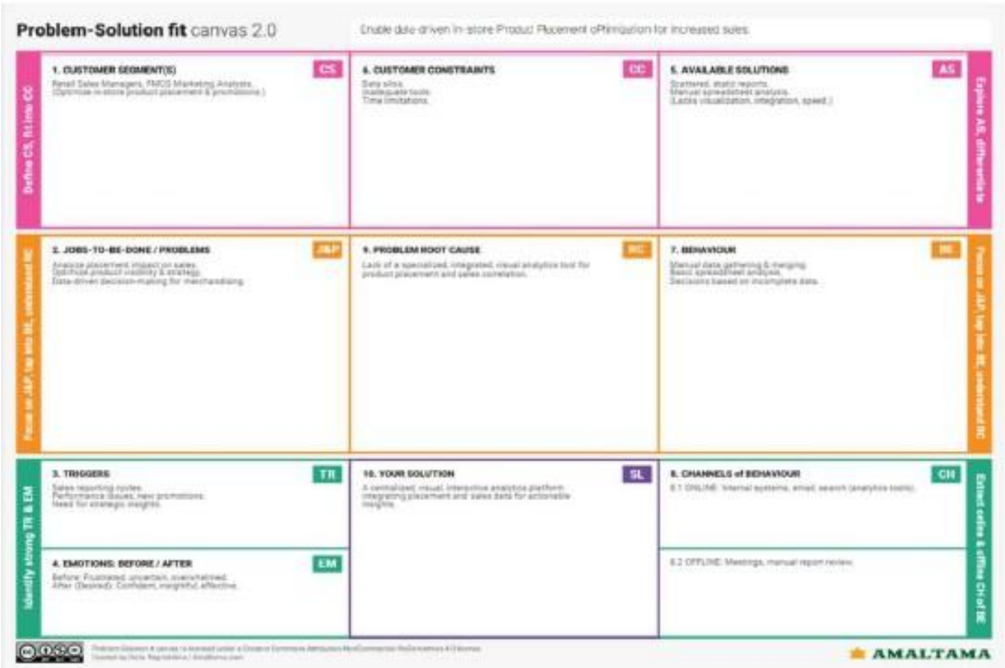
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:



4.2-Proposed Solution:

Proposed Solution Template:–

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Many businesses do not know the best location to place products in stores. Poor product placement reduces customer attention, foot traffic, and sales. The project aims to identify the best product placement strategy using data analysis.
2.	Idea / Solution description	Analyze sales, foot traffic, customer ratings, and product categories using Tableau. Create dashboards and visualizations to identify the most effective product placement strategies and provide recommendations to improve sales.
3.	Novelty / Uniqueness	The solution combines multiple factors such as sales volume, customer behavior, foot traffic, and product positioning into one interactive dashboard, helping businesses make data-driven decisions.
4.	Social Impact / Customer Satisfaction	Better product placement makes shopping easier for customers, improves their shopping experience, and helps businesses increase customer satisfaction and sales.
5.	Business Model (Revenue Model)	etail stores and supermarkets can use the dashboard to improve product placement. The solution can be offered as a consulting service or through a subscription-based analytics platform.
6.	Scalability of the Solution	The solution can be used by supermarkets, malls, grocery stores, pharmacies, and e-commerce businesses. It can be expanded to multiple stores and different locations.

4.3-Solution Architecture:

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

5- PROJECT PLANNING s SCHEDULING:

Project planning begins by defining the project objectives and business requirements. Relevant data sources are identified, collected, and prepared for analysis.A timeline is created for data cleaning, dashboard development, testing, and deployment.Roles, responsibilities, and resources are allocated to ensure smooth project execution.The project is reviewed regularly to ensure it meets deadlines and delivers meaningful insights.

5.1-Project Planning:

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a team lead, I have gather the data that is required to do the project.	2	High	Chintakunta Mounesh
Sprint-1		USN-2	As a team lead, I have already gather the data.so,I also performed loading tha data.	1	Medium	Chintakunta Mounesh
Sprint-1	Data Preparation	USN-3	As a member, I have handle the missing values.	3	Low	Kuruva Siva
Sprint-1		USN-4	As a member, I have created fields as per the requirement.	3	Medium	Kuruva Siva
Sprint-1		USN-5	As a member, I have handled the inconsistency in the data.	3	High	Kuruva Siva
Sprint-2	Data Visualization	USN-6	As a team lead, I have created different data visualizations like bar chart.	2	Medium	Chintakunta Mounesh
Sprint-2		USN-7	As a team lead, I have created different data visualizations like pie chart.	2	Medium	Chintakunta Mounesh
Sprint-2		USN-8	As a team lead, I have created different data visualizations like line chart.	2	Medium	Chintakunta Mounesh

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Data Visualization	USN-9	As a team lead, I have created different data visualizations like map,bar chart,etc..,	5	Medium	Chintakunta Mounesh
Sprint-2	Dashboard	USN-10	As a member, I have developed dashboard.	5	High	Lakshmi Thulasi Chinnappa
Sprint-2	Story	USN-11	As a member, I have developed story.	5	High	Lakshmi Thulasi Chinnappa

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Jun 2026	29 Jun 2026	12	28 Jun 2026
Sprint-2	20	6 Days	30 Jun 2026	05 Jul 2026	21	03 Jul 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

Example: $$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

AV=33/12=2.75

6- FUNCTIONAL AND PERFORMANCE TESTING:

Model Performance Testing:

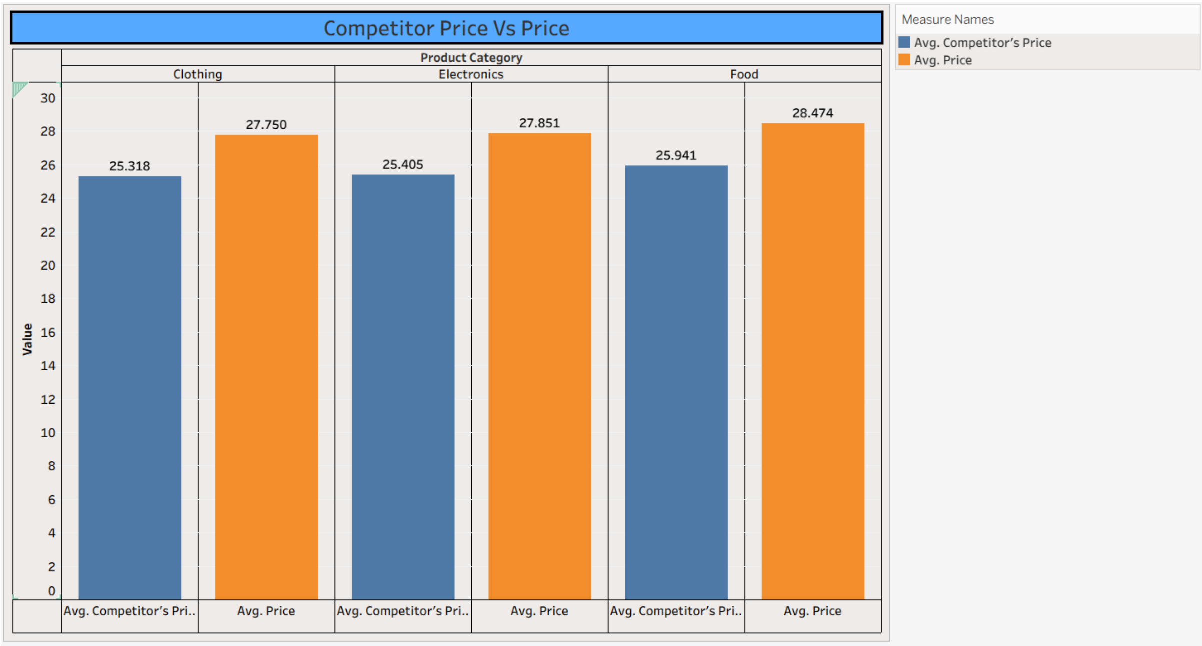
Project team shall fill the following information in model performance testing template.

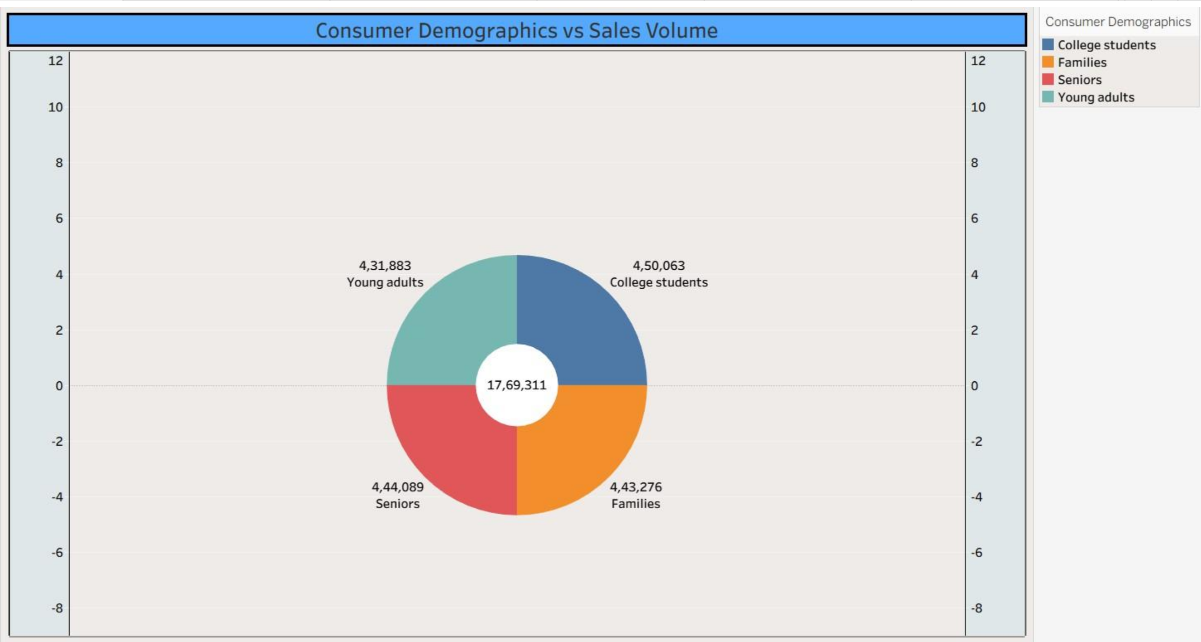
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Data of10 Rows and 1000 Coloums has rendered from an csv type file named as Product Positioning.The size of the file is 66 KB.
2.	Data Preprocessing	Handle missing values and removed Duplicate values
3.	Utilization of Filters	Measure Names,Product position , Consumer demorphics , product category
4.	Calculation fields Used	Used 1 field.It is named as zero. Syntax:0
5.	Dashboard design	No of Visualizations / Graphs – 8 Visualizations 1.Avg Sales Volume vs Product Category 2.Avg Sales Volume by Product Category by Product Position 3.Avg Sales Volume by Product Category by Season 4.Competitor Price Vs Price 5.Consumer Demographics vs Sales Volume 6.Foot Traffic by Avg Sales Volume 7.Product Category vs Price 8.Promotion of Product Category on Price and Sales Volume
6	Story Design	No of Visualizations / Graphs – 8 Visualizations 1.Avg Sales Volume vs Product Category 2.Avg Sales Volume by Product Category by Product Position 3.Avg Sales Volume by Product Category by Season 4.Competitor Price Vs Price 5.Consumer Demographics vs Sales Volume 6.Foot Traffic by Avg Sales Volume 7.Product Category vs Price 8.Promotion of Product Category on Price and Sales Volume

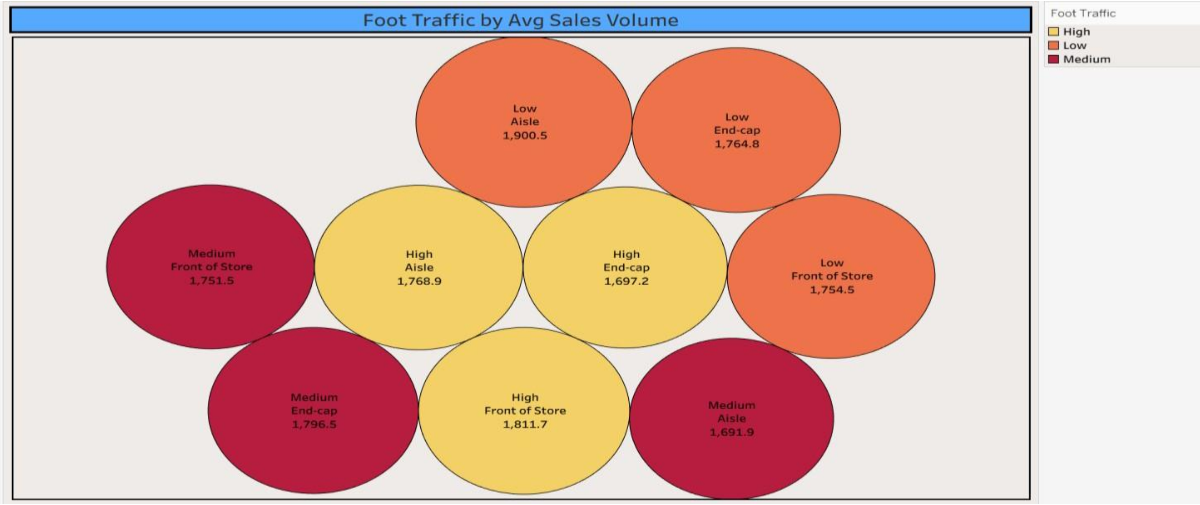
7- RESULTS:

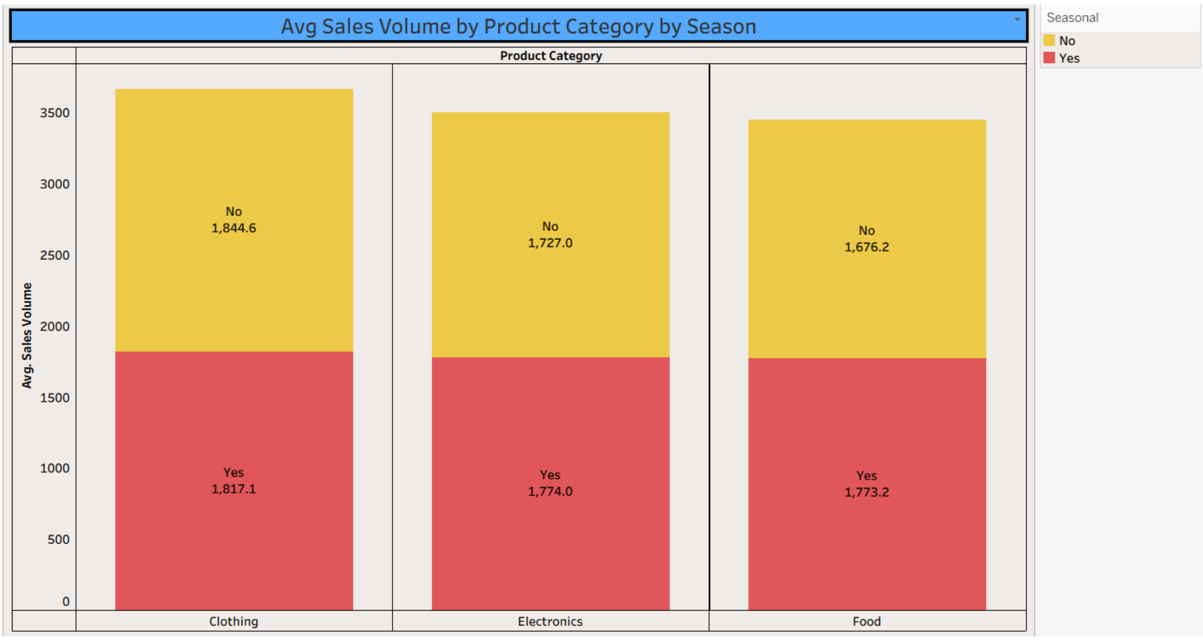
7.1-Output Screenshorts: Created

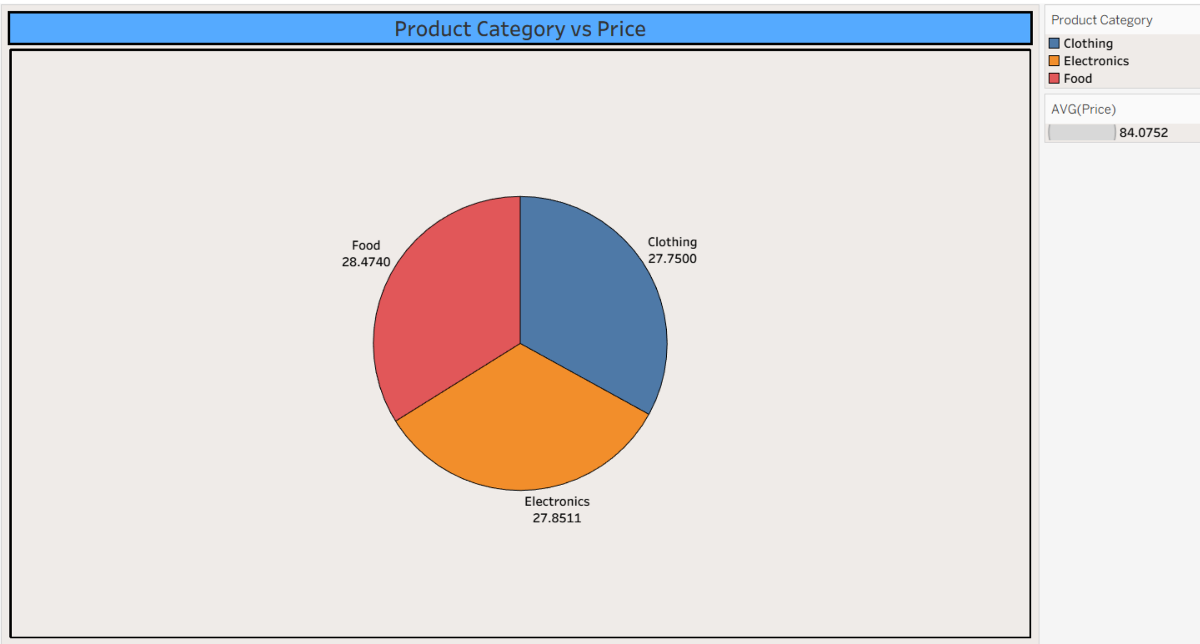
Charts:






Foot Traffic by Avg Sales Volume
Low
Aisle
1,900.5
Low
End-cap
1,764.8
Medium
Front of Store
1,751.5
High
Aisle
1,768.9
High
End-cap
1,697.2
Low
Front of Store
1,754.5
Medium
End-cap
1,796.5
High
Front of Store
1,811.7
Medium
Aisle
1,691.9
Foot Traffic
High
Low
Medium

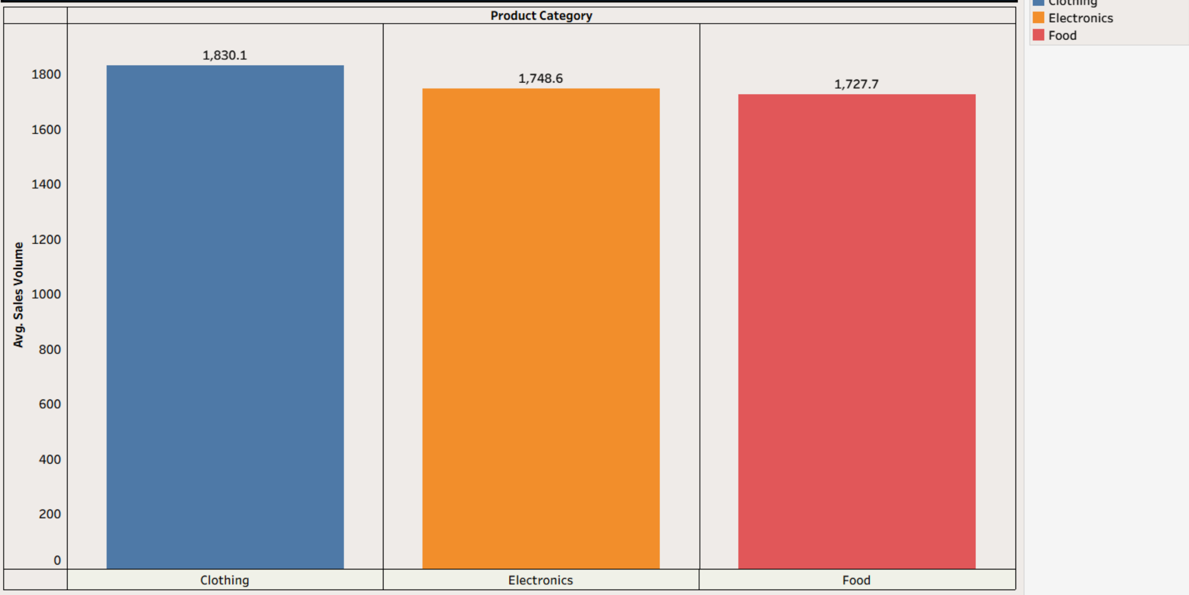

Avg Sales Volume by Product Category by Season
Product Category
Avg. Sales Volume
3500
3000
2500
2000
1500
1000
500
0
No
1,844.6
Yes
1,817.1
No
1,727.0
Yes
1,774.0
No
1,676.2
Yes
1,773.2
Clothing
Electronics
Food
Seasonal
No
Yes


Product Category vs Price
Food
28.4740
Clothing
27.7500
Electronics
27.8511
Product Category
Clothing
Electronics
Food
AVG(Price)
84.0752

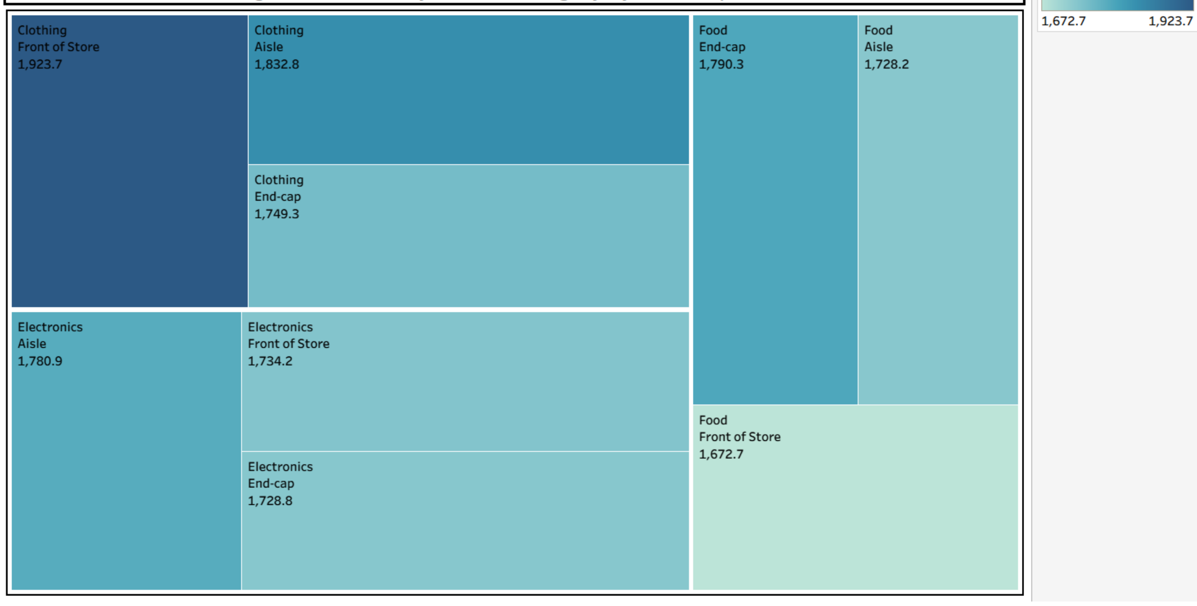
Promotion of Product Category on Price and Sales Volume

Promotion	Product Category	Avg. Price	Avg. Sales Volume
No	Clothing	27	1,869
	Electronics	27	1,726
	Food	28	1,677
Yes	Clothing	29	1,781
	Electronics	29	1,773
	Food	29	1,782

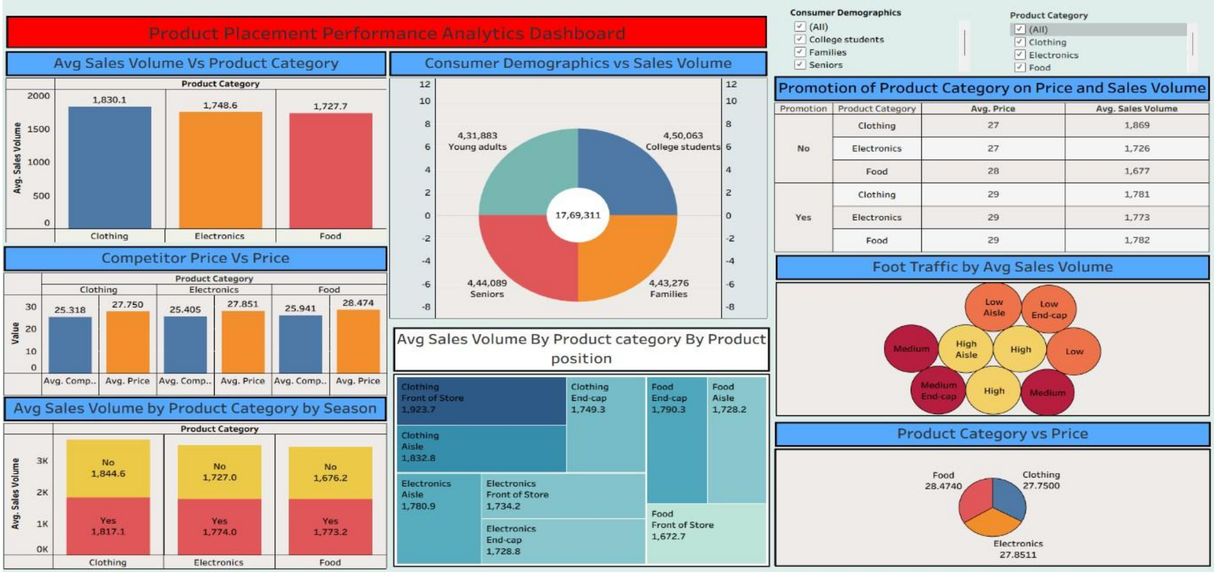
Avg Sales Volume Vs Product Category



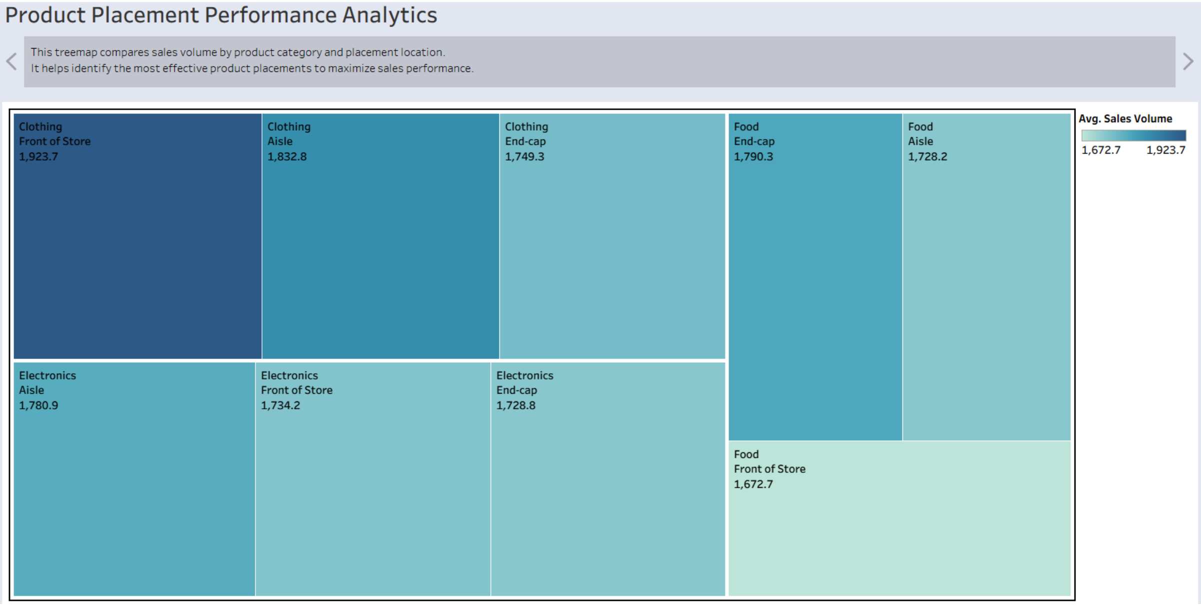
Avg Sales Volume By Product category By Product position

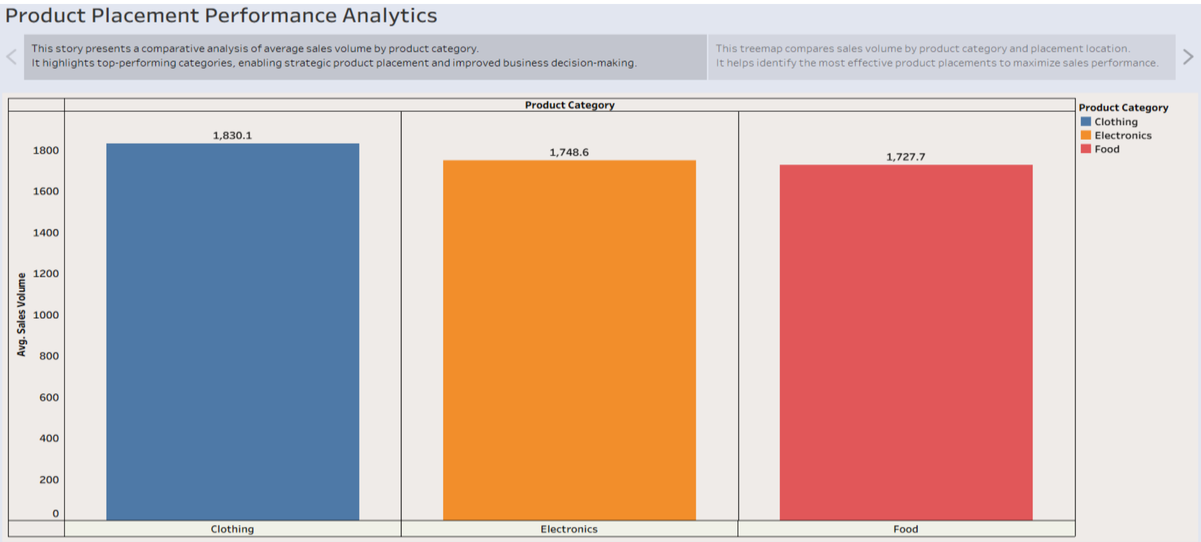


Dashboard:



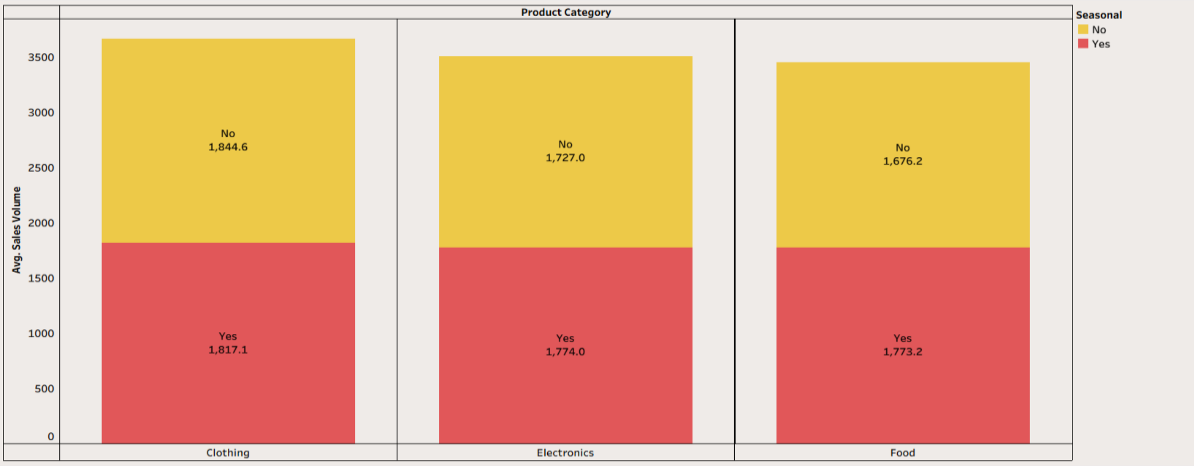
Stories:





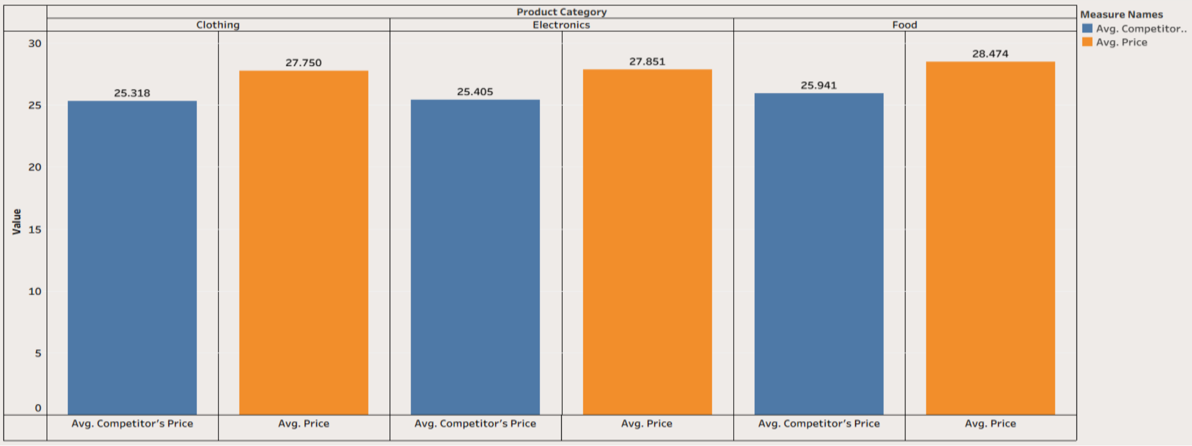
Product Placement Performance Analytics

This visualization evaluates the effect of seasonality on average sales across product categories.
It demonstrates that seasonal products outperform non-seasonal products, supporting seasonal inventory and marketing strategies.



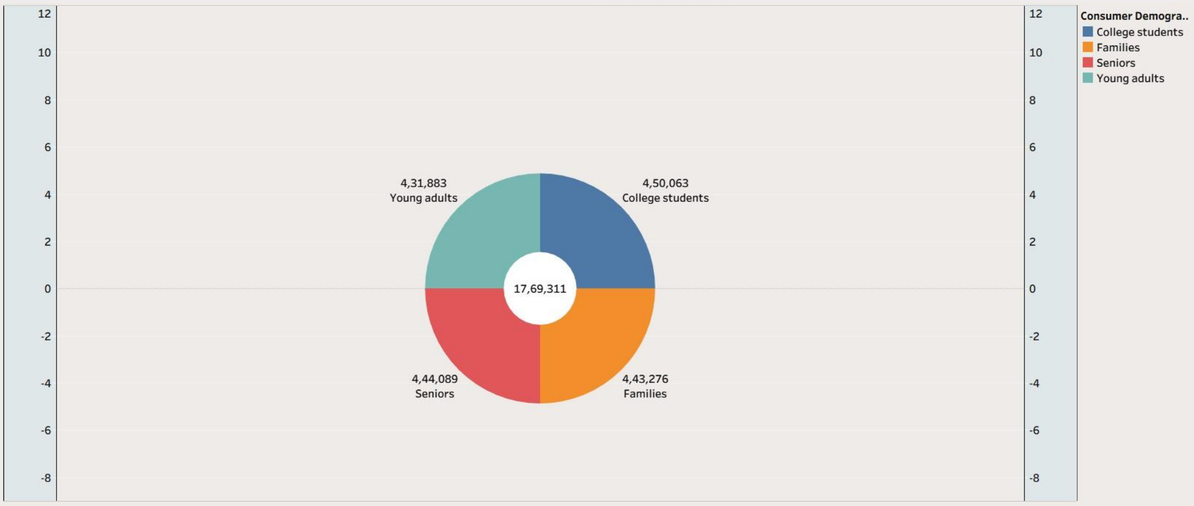
Product Placement Performance Analytics

This visualization analyzes the relationship between product prices and competitor prices across categories.
It provides insights to optimize pricing strategies while maintaining a competitive advantage.



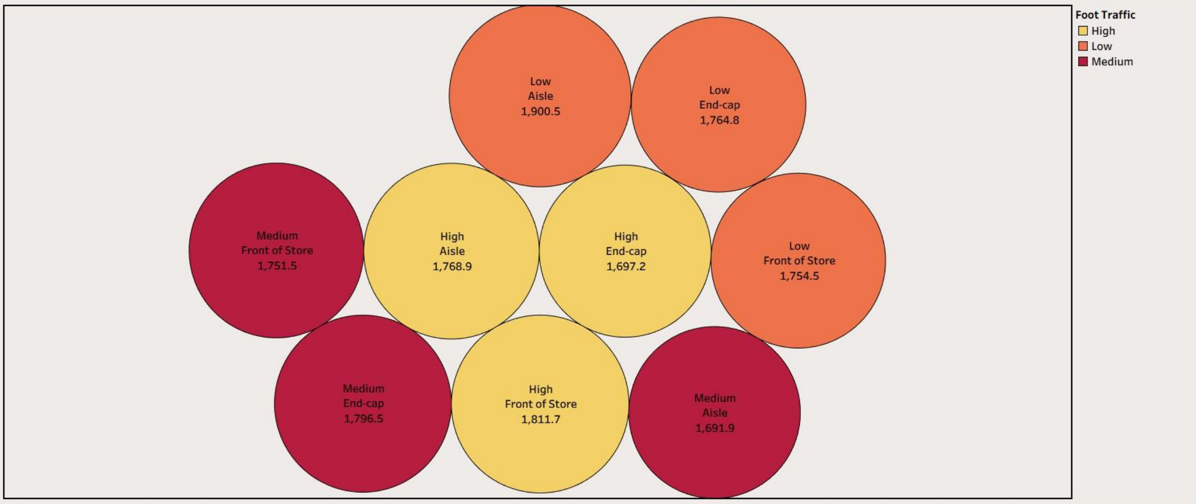
Product Placement Performance Analytics

Sales volume is balanced across consumer demographics, showing no single segment dominates the market. A broad, inclusive marketing approach can maximize overall sales performance.



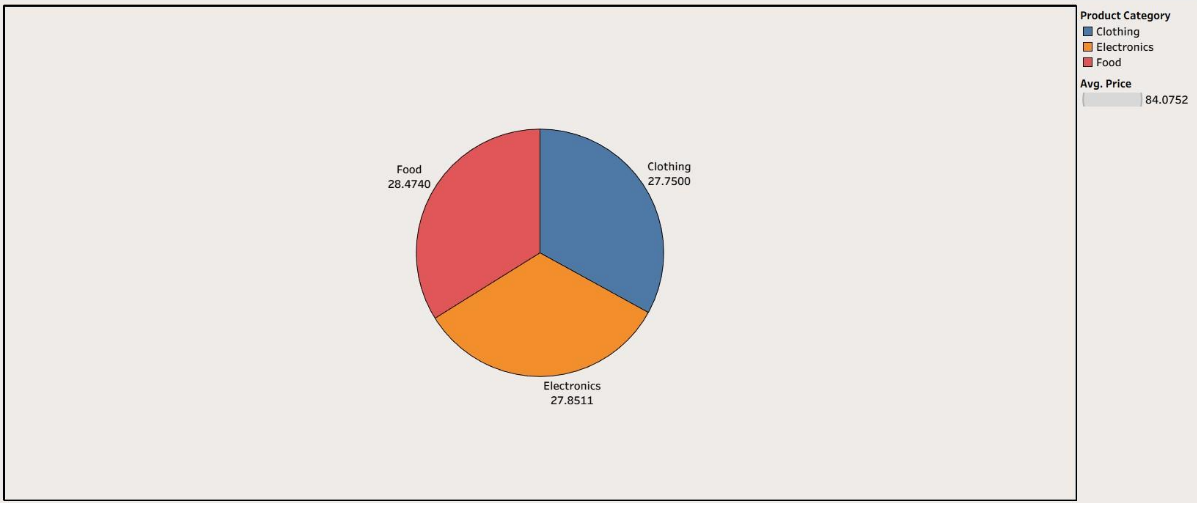
Product Placement Performance Analytics

Products placed in high-traffic areas, particularly the front of the store and aisles, generally achieve stronger average sales.
Strategic placement in high-visibility locations can maximize customer engagement and revenue



Product Placement Performance Analytics

Average product prices are consistently distributed across Clothing, Electronics, and Food, with Food having the highest average price.



Product Placement Performance Analytics

Promoted products consistently achieve higher average sales volumes than non-promoted products across all categories.
This highlights the positive impact of promotional campaigns on improving product performance.

Promotion	Product Category	Avg. Price	Avg. Sales Volume
No	Clothing	27	1,869
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	Food	28	1,677
Yes	Clothing	29	1,781
	Electronics	29	1,773
	Food	29	1,782

8.ADVANTAGESsDISADVANCETAGES

Advantages:

Creates a unique identity for the product in the market. Helps

attract the right target customers.

Increases brand recognition and customer loyalty. Provides a

competitive advantage over rival products.

Supports higher sales and profitability by highlighting product value. Improves

marketing effectiveness by delivering clear brand messages.

Disadvantages:

Requires continuous market research and analysis. Can be

expensive due to branding and marketing costs.

Incorrect positioning can confuse customers and reduce sales. Changing

customer preferences may require repositioning.

Strong competition may make it difficult to maintain a unique position. Repositioning a

product can be time-consuming and costly.

G-CONCLUSION:

Strategic product positioning helps businesses differentiate their products, attract customers, and improve sales. However, it requires ongoing research, investment, and adaptation to changing market conditions to remain effective.

The future of strategic product positioning lies in the use of AI, data analytics, and customer-centric approaches. These advancements will enable businesses to make smarter decisions, improve competitiveness, and achieve sustainable growth.

10- FUTURE SCOPE:

1. **AI-Driven Positioning:** Businesses can use artificial intelligence and predictive analytics to understand customer preferences and optimize product positioning.

2. **Personalized Marketing:** Companies can create personalized product positioning strategies based on customer behavior, demographics, and purchase history.

3. **Real-Time Market Analysis:** Advanced analytics tools will help businesses monitor market trends and adjust positioning strategies quickly.

4. Integration with Big Data: Combining data from multiple sources (sales, social media, and customer feedback) will provide deeper insights for better decision-making.

5. Expansion to Global Markets: Strategic positioning can be adapted for different regions and cultures, helping businesses enter international markets successfully.

6. Enhanced Customer Experience: Future positioning strategies will focus more on customer satisfaction, sustainability, and value creation to build long-term loyalty.

11- APPENDIX:

Github Link:

https://github.com/Narendra-0212/stragetic-product-analysis-.git

Project Demo Link:

https://drive.google.com/file/d/1MTN3kgmlK4tg1BuaNMVvy0TON3z9uVFA/view?usp=sharing